AI FOR BEGINNERS

Module 11: Risks and Limitations of AI

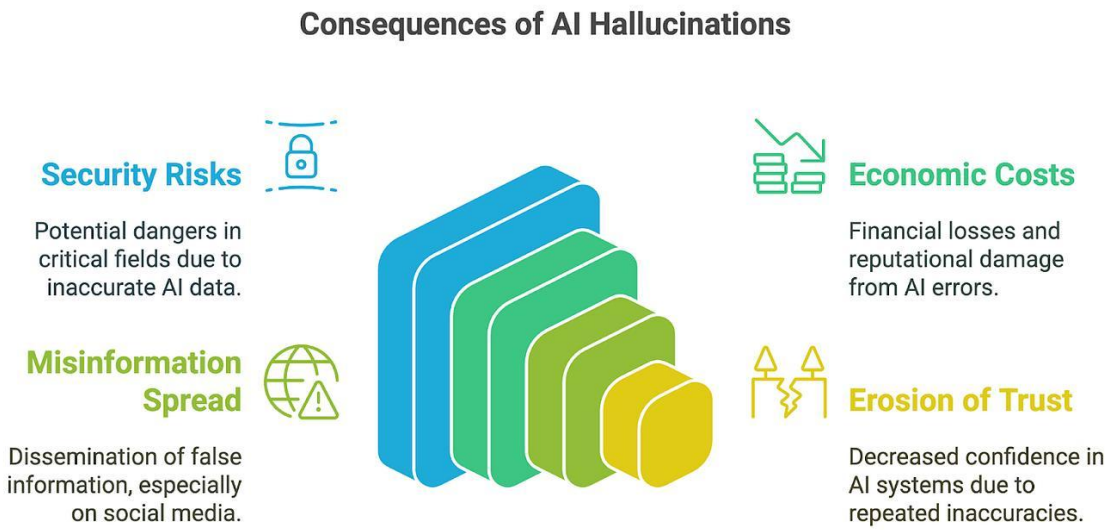
Introduction to AI Risks and Limitations

Artificial Intelligence is a powerful technology, but it is not without risks and limitations. While AI can automate tasks, improve efficiency, and provide valuable insights, it also comes with challenges that users must understand. Over-reliance on AI without awareness of its limitations can lead to incorrect decisions, ethical issues, and security risks.

Understanding these risks is essential for responsible AI usage. Users must recognize that AI systems are not perfect and require human supervision to ensure accuracy and fairness.

Simple understanding:
AI is powerful, but not always reliable

Visual Overview of AI Risks



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Common risks include:

Risk	Description
Bias	Unfair outcomes
Hallucination	Incorrect outputs

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Risk	Description
Overdependence	Excessive reliance
Security risks	Misuse of AI

Bias in AI

AI systems learn from data, and if the data is biased, the AI will also produce biased results. This can lead to unfair decisions, especially in areas like hiring, lending, and law enforcement.

Example:

- AI favoring certain groups over others
- Biased recommendations

Bias is a major concern in AI development.

AI Hallucinations

AI hallucination refers to the situation where AI generates incorrect or misleading information that appears convincing. This happens because AI predicts outputs based on patterns, not actual understanding.

Examples:

- Incorrect facts
- Fabricated information
- Misleading answers

Users must verify AI-generated content.

:Overdependence on AI

Overdependence occurs when users rely too much on AI without critical thinking. This can reduce human skills and lead to poor decision-making.

Examples:

- Relying on AI for all tasks
- Not verifying outputs

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- Ignoring human judgment

AI should assist, not replace human thinking.

Security Risks

AI can be misused for harmful purposes, including:

- Cyber attacks
- Deepfakes
- Data breaches

These risks highlight the importance of cybersecurity measures.

Lack of Transparency (Black Box Problem)

Many AI systems operate as “black boxes,” meaning their decision-making process is not easily understandable. This lack of transparency makes it difficult to trust AI decisions.

Example:

- AI gives result without explanation

This is a challenge in critical applications.

Data Dependency

AI systems depend heavily on data. Poor-quality or insufficient data can lead to inaccurate results.

Key issues:

- Incomplete data
- Outdated data
- Incorrect data

Data quality directly impacts AI performance.

Ethical Concerns

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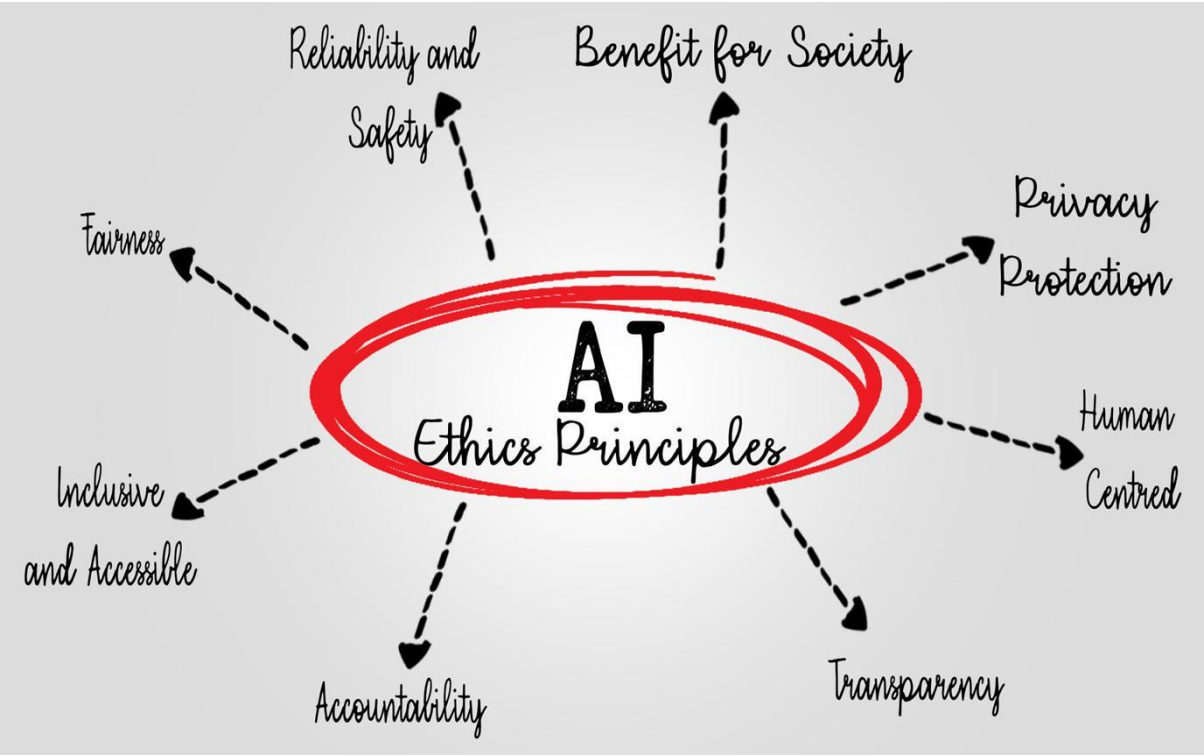

AI raises ethical questions such as:

- Privacy
- Fairness
- Accountability

Responsible AI usage is essential to address these concerns.

Job Displacement Concerns

AI automation can replace certain jobs, especially repetitive tasks. While AI also creates new opportunities, job displacement remains a concern.

Examples:

- Automation in manufacturing
- AI replacing repetitive roles

Upskilling is important to adapt.

Real-World Example

Consider AI in recruitment. If trained on biased data, the system may unfairly reject candidates based on gender or background. This shows how bias can impact real-world decisions.

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Managing AI Risks

Solution	Description
Human oversight	Monitor AI outputs
Data quality	Use clean data
Ethical guidelines	Responsible usage
Security measures	Protect systems

Best Practices for Safe AI Use

- Always verify outputs
- Avoid sharing sensitive data
- Use trusted tools
- Maintain human control

Key Takeaways

- AI has risks and limitations
- Bias and errors are common issues
- Overdependence should be avoided
- Responsible usage is essential

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